Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions required for a body to be in equilibrium.

.....

.....

.....

.....[2]

(b) Block A of mass 1.7 kg is connected by a light inextensible string to a counterweight B of mass *M* via a system of two massless and frictionless pulleys P and Q. Block A is balanced on a smooth plane inclined at 45°.

Pulley Q is a double pulley made up of two fixed wheels of different radii attached to a common axle, with the radii of the smaller and larger wheels *r* and 3*r* respectively as shown in Fig. 1.1.

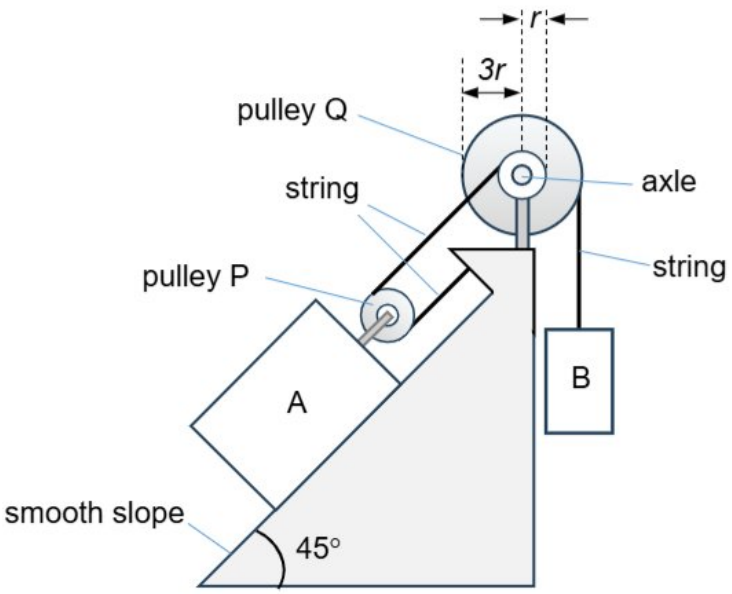


Fig.1.1

(i) Block A is in equilibrium. In Fig. 1.2, draw and label all forces acting on block A.

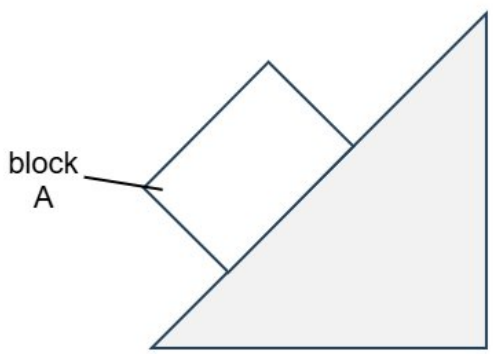


Fig.1.2

[2]

(ii) Calculate the tension in the string connected to pulley P.

tension = ... N [2]

(iii) Determine the mass *M* of counterweight B.

M = ... kg [2]

(iv) Suggest and explain a change that can be made to the pulley such that block A will accelerate up the slope.

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.....[2]

[Total: 10]